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Collateral Data Quality Challenges of IoT Sensor-Generated Data

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COLLATERAL DATA QUALITY CHALLENGES OF IOT SENSOR-GENERATED DATA

by

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ABSTRACT

Thousands of academic articles have been written about the various facets of the Internet of Things (IoT). Added to those are books of multiple flavors, conference proceedings, and a host of web-based content authored by a diverse cast of IoT community constituents. While there are many examples of successful IoT application solutions, participating technologies and how best to use them, are still relatively immature. These solutions are complex, geographically diverse, incorporate a broad spectrum of ever-evolving technologies that allow organizations to gather new data, create new value and do new things they haven't been able to do effectively before.

Despite the promise, excitement, and future thinking that leads organizations to pursue IoT projects, many of those projects have fallen short of original expectations. Too many projects fail, never making it past the pilot stage for many reasons. Authors elsewhere have written and are continuing to write about those IoT experiences, so they are not addressed in this paper. Instead, this paper looks at three critical data issues arising from the deployment of IoT solutions: 1. The magnitude of data being collected and stored from IoT solutions, and challenging if all such data is necessary; 2. How little of this data is actually being used and why; and 3. The concerns held by many users about poor data quality, what causes it, and what can be done about it. At the heart of every IoT solution are multitudes of sensors of all shapes, sizes, and purposes. It becomes apparent that sensors may be the largest source of data defects that when unresolved contribute significantly to poor solution data quality. To better understand this premise, this paper looks at the challenges inherent in sensor-generated data and possible approaches for mitigating its impact.

TABLE OF CONTENTS

ABSTRACT	ii
CHAPTER 1 INTRODUCTION	1
1.1 Thesis Research Questions	5
1.2 Summary of Chapters	6
CHAPTER 2 DATA QUALITY	7
2.1 Defining Data Quality	8
2.2 Big Data Characterizations	10
2.3 Dimensions of Data Quality	11
2.4 Metadata	16
CHAPTER 3 IoT SENSOR QUALITY CHALLENGES	18
3.1 Sensor Concerns	22
3.2 Sensor Data Characteristics	22
CHAPTER 4 IoT SOLUTION DATA QUALITY	33
4.1 The Core Value of IoT Solutions (IoTS)	33
4.2 An Example of Cost	36
CHAPTER 5 CONCLUSION AND FUTURE WORK	52
5.1 Thesis Questions and Conclusions	53

5.2	Future Work	58
5.3	Epilogue	63
	BIBLIOGRAPHY	64

CHAPTER 1

INTRODUCTION

The evolving quest for incremental organizational value is a primary driver for implementing sensor-driven solutions. Pursuit of value; new capabilities; better performance; reduced operating costs; optimized operations; better customer experience; some may not even know...Value may be perceived or manifested in so many ways.

The technology propelling this growth is the Internet of Things, or IoT, and encompasses the billions of devices worldwide connecting directly or indirectly to the Internet. These devices communicate machine to machine, passing multitudes of diverse data types and formats at widely varying speeds between them. Since the introduction of the Internet and the World Wide Web in the 1980s and 1990s, the drivers of growth have been applications, services, and data aimed at allowing humans to seek out and find information and content in which they have an interest. This activity continues to grow at compounded rates each year and remains central to the overall growth of the Internet.

The IoT is one of the core evolving technologies built around the developing universe of sensor-based devices that can measure finite physical events and conditions, convert those readings to data that is shared with other devices to make selected adjustments in the physical environment, and then store that data as prescribed so that it can be made available to authorized data consumers for their use and analysis. This data is stored in data warehouses, clouds, lakes, ponds, and "puddles," and is accumulating at astounding rates.

With the advent of increasingly smaller and more capable microchip sets in the late

1990s, industrial innovators began to focus on the development of digitally-enabled sensors and controllers to replace their long-established analog predecessors wherever they might be. While the early transitions encountered many obstacles and challenges, these became the digital ancestors of the billions of such devices that today comprise the IoT.

One of the primary defining characteristics of IoT devices is that they communicate with other like and supporting devices through machine-to-machine connections. Another defining characteristic is that each type of device is designed for a particular set of one or more functions or tasks and generates real-time data attendant to its intended purpose. The number of devices comprising the IoT is increasing at an extraordinary rate. The IoT is the market and technology supergroup of segments that include Industrial IoT, Industry 4.0, Internet of Systems, and other less pervasive variants centered around machine-to-machine (M2M), sensor, radio, audio, and video devices that generate data that traverses private and public networks. Component networks include a wide variety of power and bandwidth ranges, protocols, and transport technologies.

Through evolution and continuing innovation, the scope of what these devices can do, as well as how effectively they do it, has brought them to today's smart devices that are more intelligent, faster, cheaper, and more capable than ever. This rapid-paced evolution is expected to continue at least through the new decade. IoT devices include the sensors/actuators, the controllers they talk to, and the objects of which they are a part. Their respective physical sizes, designs, and operating configurations are tailored to their specific roles, locations, and environments in which they are intended to operate. They may be small enough to operate in finite Nanospaces, or large enough to survive in hostile industrial operating conditions. An IoT device intended to perform a specific set of functions or tasks in one particular set of circumstances may look and operate very differently than a like device in a different

set of circumstances. While some IoT devices may communicate directly to other devices connected to the Internet, most don't. In many implementations, IoT devices belong to a closed network that includes a gateway that manages that connection to the Internet. And the solution community is struggling on many fronts to manage semi-structured data across a controlled infrastructure. The challenges are likely to compound perhaps to quantum levels when heterogeneous unstructured data is added to the equation and IoT devices find their own data stream connections and routes.

The numbers are substantial. Market analyst firm IDC estimated in 2019 a population of 41.6 billion connected IoT devices generating 79.4 zettabytes (ZB) of data (79.4 trillion GB) by 2025 with a 28.7% compound annual growth rate. In 2020, that estimate was 40 ZB, about 70% of which is believed to be generally un-modeled unstructured data that includes audio, video, emails, documents, social media data, and the like. The remainder is assumed to be semi-structured data that includes nearly all traditional business, operations, and scientific research data. Increasingly more of this data is stored in data warehouses, data lakes, and cloud, fog, edge, and local data centers considered constituent elements of cloud computing and big data. It can be expensive to store data in these facilities and argued that current models might not sustain the projected growth either.

Every transport of data across the network also has a cost that is usually a function of the time-of-day, distance, bandwidth used, latency, and volume of data being transported. Like storage, network costs and capacities may not accommodate the many new network sources, traffic volumes, and speeds needed to meet this projected demand. Among the many challenges this may represent is identifying possible options for preempting the need to move and store material portions of this data through expanded use of localized machine learning and data mining methods to discern better what needs to be sent and what does not.

The number of new IoT devices and the data they will create and add to the

existing data ecosystem in just the next four years has the potential to overwhelm that ecosystem. Just doing more of what the industry has been doing will probably not be enough manage it all more effectively. New thinking, strategies, and methods will be needed to make even a small positive impact.[38] [40] Nevertheless, I anticipate that the principles and practices proposed here may be able to contribute in some way to this critical discussion around how industry participants may more effectively cope with the challenges and issues that this extraordinary growth in the number of IoT devices, their location, and resulting data volumes the next few years are expected to bring.

Curiously, only a small percentage of this data is actually being used.[21] [48] Why? Too much data across multiple platforms. It is often difficult to learn what data might be available, where it might be, how accessible it might be, and what kind of shape it may be for the needed purpose. More often than not, much effort is needed to cleanse and prepare data before it can be used. In many cases, every potential user usually has to duplicate this effort.[55] Data quality is perceived to be disappointing and falls below what is considered acceptable accuracy and reliability to glean the intended value.[51] In 2016, IBM estimated that the cost of poor data quality to US organizations was \$3.1 trillion. Gartner research estimated an average annual organizational impact of \$9.7 million. This is nontrivial and is shared by nearly all in wasted effort, bad organization decisions based on bad data, and all of the rework required to compensate.

This is about data quality. To deliver the value needed from this evolving realm of technology-based solutions must meet high standards of data quality.[36] Sensor-driven solutions are complicated and are usually expensive to engineer, build, deploy and support. Worthwhile if the perceived value exceeds the cost and effort. While some solutions are great examples of the value realization that is possible, many are falling short for a variety of reasons.

While many definitions abound, the fundamental premise about the quality of data is that it is accurate, reliable, and trustworthy; meets the expectations of those who might use it, and; is fit for its intended purpose(s). There will be more impacting elements in the "IoT-datasphere," between the moment data is created and when and where it ultimately becomes available to be consumed for its intended purposes. Data can be improperly populated, lost, corrupted, misrouted, misunderstood, or misused throughout its journey. Data issues not being effectively addressed today will only become exacerbated with more sensors in more locations generating more data.

1.1 THESIS RESEARCH QUESTIONS

So, the quantity of installed sensors are projected to grow at double-digit rates for the foreseeable future. As it goes online, each of those sensors will begin a lifetime of service generating high volumes of data. And given that each will be operating in its own mix of environmental contaminants and conditions that will impact the quality of the data it generates. These thought-provoking facts lead to five questions that this thesis document will attempt to address.

1. Given the increasing magnitude and velocity of sensor-generated data being produced, transported, and stored, the question must be asked: "is all this data necessary?"
2. How much of this data is actually being used?
3. What strategies can be applied to materially reduce data volumes and their attendant resource demands and costs?
4. What data quality challenges are a result of environmental and operational influences on sensors supporting IoT

5. What methods can be introduced or expanded upon to improve IoT data quality?

1.2 SUMMARY OF CHAPTERS

IOT solutions have many parts and are complicated to design, configure, implement, and manage. Sensors are at the heart of these solutions but are significant contributors in and of themselves to the data quality problem. This paper provides a closer look at sensors and the attributes that can compromise data quality, the environment and circumstances wherein they must operate, and solution actions that can be taken to address the issues that arise.

The remainder of this paper is organized in chapters. The following outlines the content of each of the chapters. In Chapter 2, the more salient literature supporting the key topics of this paper are discussed. In Chapter 3, key aspects of data quality are discussed. In Chapter 4, sensors and their inherent data quality challenges are explored. In Chapter 5, observations, conclusions, and future opportunities are presented.

CHAPTER 2

DATA QUALITY

In the digital world today, data drives everything. More and more, data is becoming the end-game. Data is the fuel that enables new value in many forms and better operations optimization and efficiency. Data provides expanding opportunities to explore, discover, and gain new understandings heretofore not possible about our world, our people, our customers, and relationships between them. But data is only truly useful if it is high quality which will be discussed in the next sections.[37] [42] [45]

Virtually everything important regarding data value is in its representation and analysis outcomes. Critical to useful analytical outcomes is data accuracy and high data quality. Because of the very high volumes of sensor data arising from the IoT; machine learning, advanced neural networks, and statistical modeling are used extensively to cleanse, classify, and understand trends, patterns, and even the otherwise hidden subtleties of the data. Unfortunately, even small errors, omissions, and other data defects can cause these methods to yield false and/or biased results. Result defects can become even more troublesome when further analysis using these methods can compound the effects of those defects. Decisions based on those results could be costly and ill-advised.[55] [53] [54] [66] [50]

The focus of this paper is about the nature of sensors supporting the Internet of Things, the environments in which they operate, and approaches for overcoming or mitigating some of the data quality weaknesses they represent. This section is intended to provide an overview of data quality and to set a context for sensor-related

issues and considerations for mitigating them to the extent that circumstances permit.

2.1 DEFINING DATA QUALITY

While much has been written about data quality, there is no clear agreement about what it is, how it should be measured, and how it should be managed. The perspective of what data quality is varies by context and the interests of its primary stakeholders. Most agree that data: 1. should be of sufficient value to justify the expense and effort to collect it; 2. should deliver what it is expected to represent; and 3. should be of sufficiently good quality that it is readily usable and trustworthy by all interested consumers. While many efforts are underway to reach an agreement all can adopt, as of this writing, there is still no universally accepted standard for defining data quality, but there is a significant degree of commonality of concepts to which nearly all authors about the subject seem to agree.

The concept of data quality actually describes two sides of the same coin. One side is about supply, defining the manner in which a physical entity or event will be represented in digital form. Here, the term data quality describes the conceptual framework around how well data attributes have been defined and captured to reliably portray a physical entity or event it is all intended to represent at a particular moment in time. When every attribute and data characteristic accurately and completely represents that physical entity or event as intended, the data is deemed to be of high quality and can be used confidently as a digital proxy for that entity at the particular moment of time. In contrast, data is deemed to be of lower quality as the data is increasingly less able to serve as that reliable digital proxy for that entity or event due to error, missing or incorrect values, corruption, and so on.[7]

The other side of the coin is about demand. Here, the term, data quality, asserts the degree to which a data consumer can confidently use this particular data to satisfy specific needs and requirements—that it is fit to meet their purpose. Data deemed to

be higher quality should meet their purpose more effectively than lower quality data that as it degrades, at some point will be unable to meet their purpose.

A serious point of tension that is often ignored or unrecognized in this assessment of fitness for purpose, is the presumption by many that what is defined and provided on the supply side is fully consistent with requirements and needs of all potential data consumers and stakeholders representing the demand side. That is, that they all are happy with what is being produced. This is very often NOT the case. In too many cases, decisions are made around what the supply-side data product is going to contain, how it will function, what data validation and formatting rules will be used, and what might be stored in the metadata file, all without ever talking with their community of potential or probable consumers and stakeholders. Meanwhile, members of that community have requirements and needs that they can only assume will be met in the data produced. And they are very often disappointed.

So, the story around data quality is very much like the story of the Ford Edsel in the late 1950s. At that time, Ford Motor Company was looking for a new breakthrough model that would set the company apart from the rest of the industry and on a new path of market success. Ford engineers went to great lengths to develop a car with sleek lines, plenty of power, new features, and quality parts and craftsmanship. Ford invested considerable sums in retooling their production lines and parts inventories in preparation for the Edsel's market and production launch. Unfortunately, sales of the new Edsel were dismal, far below even their worst expectations. They did not understand what went wrong. Here clearly was a well-engineered automobile, that was roomy, comfortable, powerful, and demonstrating the finest craftsmanship. Why weren't people buying it? Fundamentally, the car was ugly and hit the market just as the industry was moving away from big heavy cars in favor of slightly smaller, lighter, sportier, and less expensive models. No one wanted it. They wanted something very different. So it is with data quality. It only is achieved when the supply-side data

product delivers effectively what the demand-side community wants in the manner that they need it and can use it.[3]

2.2 BIG DATA CHARACTERIZATIONS

Big data brings an added set of dimensions to the challenge of good data quality. A substantial portion of the data in the big data universe comes from sensors in the IoT. Laney [44] originally proposed three "V" characteristics to describe the important challenges driven by Big Data. Since then, many authors have proposed various numbers of combinations of these and other such characteristics, including several that have limited application in the discussion of IoT impacts.[58] [52] [57] [59] [67] The most common and most applicable to the quality of data accumulating from IoT solution sensors are these:

- *Value* - This is where it all starts and where it all ends. Value should serve as a constant reminder that data is only truly useful if it is worth collecting and is of sufficient good quality that it successfully fulfills its intended purpose. There is no merit in investing in an IoT solution if the data it creates isn't sufficiently useful in providing the intended value.
- *Volume* - It is the volume of data that makes it Big Data. The IoT creates enormous amounts of data, quantum levels more than ever before. The challenge this creates is that many of the more traditional methods of data quality improvement are now more difficult if not fully impracticable because they don't scale enough to accommodate the challenges of massive stores of data. Any data set that is typically larger than what a single machine can effectively handle is generally considered Big Data. In this environment data is stored and processed in blocks across multiple machines.

- *Velocity* - Velocity is the measure of how quickly data is created, processed and made available to consumers. Higher velocities compress real-time processing cycle times and influence where such processing is done. Velocity also relates to latency. Lower latency demands require processing to be done nearer to the creation point.
- *Validity* - Validity asserts that all acceptable values and formats are consistent with the rules that have been prescribed for given data records and attributes. Because of data volume and volatility, machine learning processes and algorithms, and statistical methods may be used more frequently confirm data validity.
- *Volatility* - Volatility measures how long data remains useful. This can be critical because the active life of certain data instances may be very short coming from IoT sensors due to the frequency of data observations which can be made as often as 5-10 per second (for example).
- *Variety* - Each IoT sensor type generates data in its own format and content. Many are functionally related to others for context, synchronization, and data preparation processing.
- *Veracity* - The magnitude of the data being processed and stored brings newer challenges on multiple fronts. Veracity provides the assurance that data is consistently and reliably what it is represented to be and that consumers and stakeholders can trust the facilities, infrastructure, and processes that carried the data from its point of origin to its point of consumption.

2.3 DIMENSIONS OF DATA QUALITY

Just as there are many views around varying aspects of Big Data, there are likewise many perspectives on how to characterize and measure it. Like a fine diamond, data

quality has many facets or dimensions that help describe it. Many authors have written many papers summarizing, explaining, defining, and extolling various lists of such dimensions as they might apply in various contexts. A systematic scoping review of the literature suggests that data quality dimensions and methods with real world data are not consistent in the literature, and as a result, quality assessments are challenging due to the complex and heterogeneous nature of these data.[9] [58] [16] [17] [18] [32] [33] [35] [41] [59] [64] [65] [67]

However, in 2021, the work group Data Quality of DAMA Netherlands carried out research into definitions of dimensions of data quality. It collected definitions from various sources and compared them with each other. The working group also tested the definitions against criteria derived from a standard for concepts and definitions: ISO 704. The result is a list of 60 dimensions of data quality and its definitions. Going forward, wide adoption of this definition standard should provide an authoritative reference upon which future assessments and discussions can be based, as well as providing a framework against which the plethora of author-created lists can be rationalized **dama21**.

Selected from some of those lists, the dimensions that follow appear to be most applicable to characterizing data generated by sensors and subsequently processed by IoT application solutions. Each dimension here is described as it is expected to be under the best of circumstances, or at least widely prevailing "normal" conditions. But outages occur, devices fail, errors happen, data is lost or becomes defective, communications channels get congested, and any number of like incidents can arise that can impact data in various ways. As a result, questions about "something less" need to be considered and are suggested here for further thought.[61] [28]

- *Accuracy* - Data is a faithful, reliable, and correct depiction of the actual entity (e.g. person, place, event, or object), property, or condition it is expected to represent. It should be readily verifiable through direct comparison to the real

thing or other reliable data source. *Questions:* Does every attribute need to be exactly what it is expected to be, to be useful? Which attributes cannot be inaccurate in any way? Can some attributes be less accurate than others? How inaccurate can each attribute be to be sufficiently representative and acceptably useful? What is the impact on actions taken, analyses performed, and decisions made using records containing inaccurate attributes?

- *Completeness* - All expected attributes about the entity are fully captured in the data elements specified to represent them. Each data element is populated as intended based on specific or historical knowledge establishing what should be captured, the format it should take, and the values or value ranges that best represent the entity characteristic. When defined, data attributes are usually deemed mandatory, variable, or optional and are expected to be populated accordingly. All intended data attributes representing the specific entity constitute a record. All intended records constitute a data set or file. Any number of methods may be used to determine the level of completeness and non-compliance, and to insert or replace selected values outside what is intended using historical profiles. *Questions:* Which mandatory attributes must be complete to allow the record to be used? Which mandatory attributes can be unpopulated with little to no consequence? What is the impact on actions taken, analyses performed, and decisions made using records containing important but incomplete attributes?
- *Consistency* - Each instance of a particular data element is populated using the same format, value or range of values, and validation rules as every other instance of that same data element. All of the intended data attributes for each record instance are expected to be the same. And all like records are expected to be the same within data sets or files. And further, subsequent data

sets or files containing records relating to the same entity and entity instances are expected to be the same as earlier data sets or files for the same entity and entity instances. Measures of consistency can be determined by comparing randomly selected instances of like records, and against historical profiles of that particular record type. *Questions:* Do consistency expectations apply to all attributes or only to selected ones? To what extent can patterns vary without degrading or negating the usefulness of the impacted records? To what extent might such discrepancies impact actions taken, analyses performed, and decisions made using records containing inconsistently populated attributes?

- *Integrity* - The concept that "it's all there", with no corruption at any level: no garbage or lost elements in one or more records, no garbage or lost records in a data sets or files, hashes are accurate and intact, and relationships between data elements, records and files are also fully intact and correct. *Questions:* If such issues are discovered, how might they impact actions taken, analyses performed, and decisions made? How large of a gap between the good data can needed actions tolerate, analytical computations mitigate, and decisions accommodate?
- *Reasonability* - While not as specific or clear as other data quality measures, reasonability questions if the depicted data and records make sense. Are the values and relationships what they should be? Do the patterns seen in the data fit similar patterns derived from historical data? Such measures tend to be subjective are contextual based upon the nature of what the data represents. *Questions:* What are the boundaries of reason beyond which data can't or shouldn't be used? How might those boundaries be defined?
- *Timeliness* - The measurement of data timeliness must be considered within the context of its volatility, or rate of change. Data that changes infrequently is

considered static and is usually not subject to this dimension of measurement. Data not considered static is considered volatile. As the rate of change (rate of refresh by newer data) increases, there is less time to use the current instance. In higher volatility situations, the current data needs to be evaluated and acted upon before the next instance arrives. There must be time between the two instances to allow this. To do so, the time between when it is created and the time it becomes available to be used (latency) must be as short as possible.

Questions: What is the impact on operational and analytical continuity if the timely regular flow of particular data streams are disrupted? What is the impact of too much data at one time? What is the impact of large gaps? What is the impact of streams getting out of sequence? What is the impact of flows slower or faster than expected?

- *Validity* - Essential to both consistency and reasonability, validity measures the extent to which data attributes have the values they are expected to have, are formatted correctly, and have the level of precision specified. Validity rules are specific to particular data attributes. Certain attributes are assigned values matching entries in lists of acceptable choices specified for that attribute wherever it may be used. Value choices may or may not be driven by values assigned to other attributes using specified conditional rules. For example, attribute A may be assigned a value within the integer range of 0-5. Attribute B is conditionally assigned a textual value selected from a table based on the value in attribute A. Other examples include ensuring that: a textual value representing a phone number is formatted a specified way; a numeric value representing a unit price is formatted and computed to three decimal places; or a numeric value representing atmospheric temperature is shown as degrees Fahrenheit formatted to one tenth of a degree and within the range +/-150 degrees. *Questions:* Which attributes must follow their respective formatting and validation rules

without exception to be acceptable and useful? Which attributes can be allowed to have values that violate formatting and validation rules? What is the impact on actions taken, analyses performed, and decisions made using records containing attributes with validation or formatting rule violations?

2.4 METADATA

In 2015, McKinsey Consultants estimated that data consumers spend 65% or more of their time, trying to find the data they need, discovering where the data has come from, learning about what issues have been identified since the data was created, assessing what treatment has been applied to the data to this point, and determining what else may need to be done to the data set or file to get it to a state where they can actually use it.

Good metadata is a critical component of good data quality, particularly for Big Data files and data sets originating from sensor-generated data streams in the Internet of Things. Metadata should be a close companion to every data set, file and stream as it is created or updated. It should be subject to a similar level of quality standards as the data it describes. Good metadata should be able to tell an interested consumer just about anything that needs to be known about the data set or file. In IoT world, metadata is particularly essential to track the journey of each new data stream as it is processed and ingested into the Big Data data stores where it will come to rest while it awaits further processing or analysis by stakeholder data consumers. The goals of the IoT technosphere is for all important aspects of any architecture and application solution be automated to provide humanless assessments, controls and event information through machine-machine interaction.

Like the data itself, comparable standards and practices should be in place for establishing and maintaining metadata. Event journaling and core documentation creation should be embedded in the data set, file, or stream metadata file at ap-

propriate action and control points throughout application solution coding to ensure that necessary information is included every step of the way and easily accessed by every user authorized to learn about and access it.

CHAPTER 3

IoT SENSOR QUALITY CHALLENGES

There are IoT devices of every shape, size, technology, purpose, and environment. Functionally, they can be grouped into a few general categories that include sensors, actuators, and controllers.

3.0.1 SENSORS

The role of the sensor is to monitor and capture data related to its intended purpose, and pass it along to other IoT devices, or place it directly on the Internet for downstream processing. They may be used singularly or in combination to provide more robust perceptions of particular situations at specific moments in time. Typically, sensors have passed their data to connected controllers for evaluation and indicated action. Increasingly, sensors are being built with their own limited computational and data evaluation capabilities. This allows them to operate more autonomously and act more quickly when circumstances suggest. For example, a sensor may determine that ambient light levels have decreased below an established threshold, and then acts to turn lights on. It may also interact with proximate motion sensors and clocking sensors to determine when or if those lights should be turned off.

There are several categories of sensors that include the following:

- *Mechanical sensors* measure both static and dynamic forces of movement in the physical world. Examples include accelerometers, gyroscopes, inclinometers, scales, and touch.

- *Geospatial sensors* ascertain where objects are in physical space including altitude and orientation, and are frequently associated with many types of mechanical and environmental sensors. These often are considered the core of a wide variety of sensor packages used by many government agencies, scientific research, and higher education for their observations of land, water, and, air conditions around the world.
- *Environmental sensors* monitor physical conditions proximate to their respective location. Examples include temperature, humidity, pressure, light, sound, motion, proximity, altitude, attitude, direction, location, and imaging. In many respects, some of these may be used singularly or in combination to emulate the human senses of sight, sound, and sensations.
- *Chemical sensors* detect the presence or absence of targeted chemical and biological substances and conditions. In many ways, they emulate the human senses of taste and smell, but to a much greater degree. Examples include air and water quality, smoke, gas detectors, and substance detection, to name a few.
- *Medical sensors* are all designed to monitor and collect data regarding the many aspects of the human body. Examples include heart rate, pulse, temperature, blood glucose, and oxygenation level, neural activity and feedback, and respiration rate.
- *Radio Frequency Identification Devices (RFID)* are designed to enable asset identification and movement. There are two classes of devices: chip-based tags containing stored static data and a radio antenna, and readers that receive the tag radio signal and captures and stores the resident data from the tag. The reader is considered the sensor and tags are what they sense. There are

several tag types and designs that are specific to their intended use, location, and environmental conditions.

Major Category	Type	Models & Configs
Audio	Acoustics, Microphones	1000+
Biometric	Distance, Oximeters, EKG, Heartrate	75+
Capacitive Touch	Buttons and Pads	250+
Electrical	AC&DC Current, Conductivity, Consumption, Transducers, Transformers, Voltage	2200+
Encoders		7100+
Environmental	Air Quality: Dust, Gas, Chemical, Ionization, Particle, Pyroelectric Detector, Humidity, Pressure, Smoke Detectors, Other Thermodynamic Sensors	1000+
Flow	Air, Gas, and Liquid Meters and Sensors	300+
Force	Load, shock, and Transducers	250+
Linear Displacement		550+
Liquid Level		1450+
Magnetic		3400+
Motion & Position	Accelerometers, Distance, Gyroscopes, Inertial Measurement Units, Inclinometers, Potentiometers, LIDAR, Radar, Scales, Speed, Tilt, Touch, Vibration, Other Geospatial Sensors	3600+
Optical	Ambient Light, Fiber Optic, Image, Infrared Detectors, Optical Switches, Photo Detectors, Photodiodes, Photoelectric, Phototransistors	14000+
Pressure		19400+
Proximity	Capacitive, Inductive, Magnetic, Optical, and Ultrasonic	19000+
Temperature	Temperature, Thermostats, and Thermistors	13700+

Table 1: Selected Examples of Sensors Used in IoT Solutions ¹

Table 1 is included to portray the broad selection of the wide variety of sensors being deployed in embedded IoT systems solutions at this time in 2021. The numbers in the right column include both the individual model of each sensor and the established configuration options that model sensor can be assigned. It is quite common

¹Source: Mouser Electronics ([mouser.com/Sensors/](https://www.mouser.com/Sensors/))

for a sensor to have as many as twenty or more configuration options. Many just have one. There are several hundred manufacturers of sensors and related components.

3.0.2 ACTUATORS

The role of the actuator is to initiate a specific change in the environment based on data coming from sensors, either directly or through associated controller(s). Examples of such changes might include: turning something on or off, increasing or decreasing speeds, flows, pressures, or temperatures. In many cases, actuators may create and insert data into the network indicating what actions they initiate and that it was executed successfully or not. Virtually all of the characteristics of sensors apply generally to actuators as well.

3.0.3 CONTROLLERS

The role of the controller is to take data from sensors, process it, and ultimately determine if an adjusting action is needed, what actuator(s) need to make those adjustments and to what extent. The intelligence to make these assessments and determinations are usually held in the controller. However, increasingly more of the intelligence is being built into the sensors and activators, as well as incorporating inter-connectivity that enables them to talk directly with each other and handle routine, and less complex, exceptions without needing to defer farther into the network. Functionally, any proximate processor can be tasked to perform the needed sensor/actuator processing. Sensors, actuators and controllers are being integrated into integrated chip packages to reduce latency and overall power consumption. An increasing number of such chip packages are being deployed in newer solutions. Larger volumes of data, more complex analysis, and time sensitivity is likely to drive demand for these as they hit the market.

3.1 SENSOR CONCERNS

The world of sensors is vast and influence the design, definition, features and functions of IoT application solutions with variants aimed at the differences in the environments within which they will be deployed. Much can be discussed about each aspect of the chosen architecture, device selection and use, deployment specifications and so on, but they will not be discussed here.[34] [47] This paper is about how selected characteristics of sensor data and the environments within which they are implemented, impact the quality of data into and out of IoT application solutions.[66] Some of common characteristics are introduced and discussed.

3.2 SENSOR DATA CHARACTERISTICS

Sensors live in the physical world. Each has a very specific purpose and is so designed. One specific type of sensor may have many performance variants and be built using any number of form factors to fit particular needs. An increasing number of sensors are in micro-electromechanical (MEM) categories may be as small as one millimeter and are often packaged with similar sized sensors of other types.[6] Environments may be harsh and hostile towards sensors and related electronics, and can have detrimental impacts on the quality of the data they create. They are often placed in hard-to-reach locations making them difficult to support and maintain. They are built to communicate be it other sensors, actuators, local controllers, or across a variety of networks. As the key components of much larger, and often very complex application solutions within the Internet of Things, the quality and reliability of the data they generate is intrinsically critical to the overall value of the overall solution of which they are a part. But, they are vulnerable to a variety of issues that can impact that data quality and reliability.[26] [61]

Some of the more important characteristics that make this so include:

- *Low Cost* - The fundamental driver for today's sensors is having the needed fundamental functionality at the lowest possible cost. This sets the parameters within which everything else about sensor design, function, form factor, on-board processing, and storage must come together. Each of the characteristics that follow in one way or another are controlled by the need to keep costs low. This has been, and continues to be, a critical success factor for enabling the design and deployment of IoT application solutions. Because of the quantities of sensors required to meet solution objectives, manufacturers are concerned that higher per-unit sensor costs may push solution costs too high to justify and dampen deployment volumes. As will be discussed further here, manufacturers are being pressed to improve what they can provide without materially raising costs.
- *Technical Limitations* - The form factors of newer generations of sensors are getting smaller, more digital, and more varied. Lines are more often blurring between sensor, actuator, and processor functions, and more processing is being attempted on board. Previous analog functionality is being shifted to digital when it is expedient and cost effective to do so. And the migration to MEM architectures and designs is increasing. As a result, there is very limited real estate available for the actual sensor elements, memory, processor, and power source. Sensor packages growing increasingly smaller; some are now as small as a millimeter. Research is being done now on moving several selected sensor elements and packages into the nano-micro-electromechanical (NEM) range. Mass-scale deployment economics attempt to keep costs as low as possible, against the rising pressure to expand on-board functionality and computing power.

The consequence of all this is that there is limited to no available resource on most sensors to help resolve the issues that are discussed in this section. But

that may not be case going forward based on the work device manufacturers are devoting to expand on-board computing, storage, and memory, particularly in devices that are using hard-wired low-voltage power sources. This trend may extend to devices not on local power as the needed technologies mature enough to allow them to do so.

- *Power* - All sensors need power. Bad things can happen when sensor power is lost or becomes unstable. Sensors are most likely to stop functioning, or may function minimally without the level and stability of power needed to do what it is expected to do. Machines may go down due to the loss of sensor control. Data production may stop or be delayed. Many are hard-wired to low-voltage premise power feeds. A wide variety use batteries, while others use ambient light, solar cells, and various other forms of harvested energy. For example, RFID tags scavenge power from readers when activated. While newer generations of ultra-efficient, ultra-low-power batteries and power cells are in development in the labs,[60] there is much to be done to advance them to the their point of readiness for wide spread service. Power must be sufficient for the sensor to make observations, convert them to digital, develop its reading, store the results in registers in memory, format output observation data and transmit it to designated receiving nodes.

Unfortunately, premise power isn't always as stable and consistent as all would expect. Interruptions or spikes in premise power, however minute can disrupt the flow of sensor processing and can cause garbled, missing, incomplete, or inaccurate data to be captured and transmitted. Many premise-based sensors include batteries to help mitigate these negative impacts. A subset of these are designed to run from the battery to provide stable, consistent power and are kept charged by the premise power. The other subset uses batteries for backup that take over when premise power becomes unstable or unavailable.

The objective in all cases is to keep the sensor running with sufficient quality power to do its job while minimizing data defects.

Battery-powered sensors are designed to run for long periods of time with healthy batteries designed for service lives of several years with some lasting as long as ten years. Battery specification and selection is critical to the longevity of the sensor. Sensors that depend only on battery power are designed to be ultra low power and very power efficient in their operation.[39] Battery life is a function of rated power and capacity divided by the average cycle depletion rate. The cycle depletion rate is the sum of the power needed for the sensor to make, evaluate, record, and transmit the data from one observation. Each resident component in the sensor has its own power demands and affects directly the cycle depletion rate. The sensor component has a specific cycle demand on power as does the memory and processor components. Larger memories and faster processors typically have higher demands on cycle power. The communications component may demand even more. A simple wired network connection may demand minimal power while any one of the possible wireless connections could take significantly more. If the sensor includes retransmission functionality needed to assure data message delivery, power demand could rise substantially to complete the impacted observation cycles when transmit errors occur.

The frequency of a sensor's observations is a major factor in sensor longevity. A sensor that makes an observation every 500 milliseconds will deplete its power far sooner than the same sensor configured elsewhere to make observations every 15 minutes. Extreme temperatures may also impact performance and battery life. Hotter temperatures tend to increase available power but shorten battery life, while colder temperatures may impede performance but extend battery life. Excessive moisture to an exposed battery can result in rust and corrosion and cut short battery life.

Sensor design, specification, and selection must consider all of these elements and the dynamics they may bring to their deployed location, environment, and function to find the best balance between power capacity, expected demand and targeted service life. In the end, the first indication that there may be a power problem is when sensor data errors begin to elevate or when data from a particular sensor stop suddenly and don't resume after a period of time. Depending on the sensor, its function, and the nature of the data it is intended to produce, this type of failure could be critical.

- *Location* - Where a sensor is placed, how it is mounted, how it is powered, and how it is expected to communicate may all have some impact on a sensor's reliability and stability in the quality and consistency in the data it delivers. Physical installation decisions together with often widely-variable ambient environmental conditions may also impact sensor performance, data quality and device life. Dust, vibration, heat, cold, humidity, moisture, UV radiation, electromagnetic interference, and airborne chemicals are common conditions that may attack sensor elements resulting in shorter device life, premature failure, and consequently increasingly unreliable data.
- *Noise* - Generally, many types of sensors function by isolating and measuring specific attributes of signals fitting waveforms of certain frequencies and amplitudes. As sensors make their observations, they expect to capture signal patterns that fit closely to the targeted pattern. Measurement algorithms attempt to discern differences between what they capture and what they expected. Under ideal circumstances, sensors can reliably capture and measure what they are designed to do and generate and transmit good data indicating the observation's results.

However, sensors can commonly encounter noise while taking their respective

readings. Noise is defined as a random unwanted and unintended disturbance in an expected electrical signal. Noise is environmental, some is atmospheric and occurs naturally, while the rest is a product of proximate electrical, and sometimes mechanical, activity. Electromagnetic Interference (EMI) and Radio Frequency Interference (RFI) are very frequent causes. Virtually everything electrical, electronic, or radio-related generates noise in some form. Most noise is not a problem outside of specific, often narrow circumstances. It generally only becomes an issue when that noise from certain external sources intrude and interferes with the intended functional operation of a particular circuit, chip, or component—such as a sensor, actuator, processor, and the like.

Resolving noise issues is often complicated and can take time. Some patterns of distortion or signal alteration have become identifiable over time resulting in filters or algorithms to mitigate their impact. Other patterns have to be investigated using a variety of troubleshooting methods that may help isolate and identify potential contributing factors.[46] For example, some of the possible noise mitigation actions might include: the sensor may need to be moved from HVAC airflows to mitigate temperature fluctuations and convection cooling; machine vibrations may need to be dampened to reduce their effect on the sensor; sensors may need to be moved to increase distance and minimize impact; lights may need to be moved or shielded; nearby unshielded cables may need to be moved; cellphones and other radio signals may need to be taken away, repositioned, or shielded; and other like actions dictated by the environment and the nature of the effects that need to be identified and mitigated—if they can be.

Unresolved noise will usually result in missing, garbled, inaccurate, or outlier data defects that will need to be resolved downstream in the data channel.

- *Accuracy* - Probably the most common and most concerning issues with sensor-generated data are the continuing accuracy of the observations being reported by each sensor. Noise, significant temperature fluctuations, and even extended normal use can cause what is referred to as drift and happens to most sensors to one degree or another. Drift is defined as the gradual increase in the gap between what the device is measuring compared to the measurement it has been calibrated to expect. As a result, data that appears complete and accurate is actually not and becomes less reliable and useful.[5] The issue becomes: just how inaccurate is this data, how long has it been inaccurate, and at what point does it become unusable and need to be dropped? To reliably produce accurate data, sensors need to be recalibrated as needed or at least checked regularly to confirm they are providing measurements within the limits for which they have been calibrated. All this said, it is important to note that sensor-generated data is not perfect. There will always be some level of error that needs to be recognized and accommodated.

The discussion about accuracy must also include some attention to precision. Sensor precision refers to how finite a specific sensor measures what it is designed to measure. There might be hundreds of one sensor type, each with its own design and function target. For example, temperature sensors have a wide range of function targets. On one end, temperature sensors targeted to measure ambient air temperature might have a measurement range of -10 degrees to +130 degrees Fahrenheit with a precision of one-tenth of one degree. On the other end, cryogenic storage monitoring sensors could be targeted at the range of -460 degrees to -238 degrees Fahrenheit with a precision of one-thousandth of a degree. The point being that a temperature sensor isn't usually one size fits all. Within each sensor type there are often several models from various manufacturers, each with their differentiating characteristics. Sensor measurement

data is often misinterpreted downstream because it isn't clear which sensor produced it, only to find that the range and precision of the data isn't what was expected.

The explanation of the specific meaning of the data produced by a particular sensor is usually provided by the manufacturer of that sensor. It may be different from other similar sensors so it is important for that sensor's documentation to be sure it is clearly understood.

- *Duplication* - Any number of operating interruptions or delays may result in sensors duplicating data being sent downstream. Fluctuating power, communications issues, outbound sensor data transmission backlogs are all common causes of resulting duplication.
- *Connectivity* - Sensors exist solely to make periodic prescribed measurements, format the readings into data records (tuples), and send them somewhere else across one or more methods of connection. For this discussion, the nature of selected possible connection options is unimportant here, but there are potential impacting issues common to them all. Specific option characteristics, technologies, and protocols aside, their purpose is to reliably transport sensor-generated data to its designated destinations. Each is expected to be configured to have capacity sufficient to allow this data to move smoothly from its source to its intended destinations. But that does not always happen. Some of the more common issues that often arise include:
 - Power fluctuations and outages can cause data streams to stop moving data causing data to back up until the data stream path can be restored, stabilized, and the data stream backlogs cleared. If more data collects in the backlog than the stream can accommodate, data will likely be lost or corrupted.

- Each sensor is configured to make an observation at regular intervals. The minimum interval cannot be any shorter than the time required to initiate and complete one observation cycle. Within each observation cycle, the sensor generates a certain amount of data. For some sensors, that amount of data is limited while others may generate considerably more. The size of any particular sensor data stream is a product of the size of its generated observation record and the frequency of its observations. Observations need to be scheduled often enough to fulfill the motivating value requirements, but no more frequent than necessary to avoid overwhelming the data stream’s bandwidth and potentially losing or significantly delaying data.
- Every data path has an established capacity within which the data may flow. Data moves freely along that path as long as its volume and speed don’t exceed the path’s established capacity. When that capacity is reached, upstream data slows and gradually pushes the blockage further upstream creating a larger backlog. Here too, data loss or corruption becomes increasingly likely. In streams where low latency is required, such delays can cause poor or missed response to a critical real-time action request.[8]
- *Fusion* - In many cases, the data from one sensor is not enough to be useful or actionable. Data from multiple proximate sensors are often needed to provide better context and meaning.[2] [4] [43] [49] For example, temperature measured at sea level may be quite different than temperature measured at the top of a 300 foot tower a short distance away. Without further information, explaining the difference between the two proximate readings is difficult. Adding altitude sensor data at each location helps show why a difference may exist. To take this further, temperature is usually not that useful without also knowing the relative humidity. So, to be useful in a broader context, sensor data represent-

ing temperature, relative humidity, and altitude need to be brought together to compute adjusted data values that may be more meaningful. This process is referred to as sensor data fusion. In this process, the GPS (or other) location, and date and time must be incorporated to complete the record(s) to be transmitted.[63] [56] [62]

For sensor networks that have been deployed for several years, it is the responsibility of the solution provider/developer to establish how all of these elements are to be brought together to create the contextualized data instance(s) and transmit them. Where standardized requirements can be established, manufacturers are developing integrated multi-sensor packages that accomplish this fusion on single components. There are some challenges arising out of this fusion that may also impact data quality.

- Observation synchronization is critical to fusion success. In established deployments, participating sensors may not be close enough together to allow the intended fusion processing to be done within the observation interval established for them. This would require this fusion to be done further down stream. Deployments incorporating integrated multi-sensor packages mitigate this problem and are likely to be more suitable for real-time, low-latency requirements.
- Just one of a participating group of sensors that gets out of sync or is delayed in providing its data, can compromise the ability of the fusion process to keep up with the steady flow of observation data from all participating sensors. As has been discussed in other circumstances, delays in process flows will cause data to back up and are likely to result in data loss and corruption. Observation cycle interval timing needs to be adjusted to minimize the frequency of such issues.

- Date and time synchronization between related sensors is essential to the ability to correlate and draw meaning from their observation readings. Shorter observation intervals require this more than with longer intervals. When intervals are shorter, the time between observations is less and tolerance for incongruity between them decreases and increases the possibility of error between them. At some point, the incongruity of times will create confusion over to which observation in a series the outlying sensor’s data belongs. This could trigger a potential duplicate resolution issue with the collective observation group immediately before or after the observation group to which it should belong, and, will likely create a missing data situation for the observation group to which it actually belongs.[30] [31] [29]

CHAPTER 4

IoT SOLUTION DATA QUALITY

4.1 THE CORE VALUE OF IoT SOLUTIONS (IoTS)

It is easy to become overwhelmed with, not only the vast volumes of IoT devices, but the seemingly endless and creative things they do, how they do it, and how they work together to achieve their collective missions. None of these exist for their own sake but for the data they create and curate and the value that data represents for the organizations that deploy it in the first place. Each deploying organization places a value on the role each device plays and the data produced. For some, the value is episodic or event-related providing some level of understanding about them. For others, it is gaining a more timely understanding of operational norms and as much as they can about exceptions or anomalies. These understandings are derived from appropriately timed and scaled analytical processes embedded in data value life cycles. From these understandings, their respective aims are to save money, improve efficiency, optimize operations, learn nuances of their business they didn't know before, and a host of other reasons and motivations. This is why they do all this.

The core value of IoT devices, and the ecosystems within which they perform, is how they are used to solve a myriad of problems, enable greater value, and fulfill mission-critical functions in a wide variety of ways. As sensor electronics and processing capabilities improve, more sophisticated and smarter applications and their constituent solutions are being developed and deployed for the new value they are

intended to provide. They become then, the foundations for future innovations.[1]
Some examples includes the following.

- Factory floor automation: Nearly every aspect of shop floors, production lines, refinery systems, materials provisioning, and a host of robotic devices are all targeted for sensing or adjustments by IoT. Line optimization, waste reduction, product quality, energy conservation, cost efficiency, security, and product tracking are all in the sweet spot for this space.
- Smart cities: Progressive cities are implementing a wide variety of IoT to improve their services to their public. These include intelligent parking, traffic flow analytics, congestion management, smart traffic lights, accident detection and response, emergency response management, security and law enforcement, utility meter reading, smart building management, and clean energy management.
- Smart transportation: Automobiles and trucks have embedded sensors, actuators, and computers onboard since the 1970s. They have continued to evolve to the point where virtually every operating system in our vehicles are monitored and controlled. Today, innovation is pushing towards vehicle autonomy and route auto-navigation. This will require sensor-bedded roads, among other things. Included also are the developing interests in smart delivery drones.
- Energy management: Throughout the power distribution chain, sensors and actuators are used to manage grid health, flow levels, threshold tolerances, supply and demand instrumentation, route selection, system security, and fault detection throughout the distribution network. Customers look to find ways to reduce their energy consumption by monitoring those elements that are drawing power and don't need to be and taking action to shut them down, particularly in factories, refineries, and office buildings.

- Environment management: Just about everything in the physical world around us can be measured, monitored, and in some cases, adjusted to help humankind in its stewardship of the world around us. Examples include air and water quality, chemistry, temperature, rainfall, humidity, salinity, and movement. Some extraordinary measures might include water levels, the presence of toxins, the absence of important materials, and related conditions that may be outside expected norms. They can be deployed for physical, chemical, and biological applications.
- Smart agriculture: An important special class of environmental applications focused particularly on the growth of crops and getting them to market. These can look at soil temperature, moisture levels, acidity, key nutrient levels, etc. They can also monitor key factors for individual crops to gauge general readiness for harvest, as well as the presence of blight and other conditions that may be detrimental to crop quality.
- Health and fitness: There is a myriad of opportunities to enrich medical and health care services, including monitoring and reporting bodily functions such as heart rates, pulse rates, breathing, blood oxygenation, and bodily activity levels. The most comprehensive of these are likely to be found in hospitals, but some may be worn by individuals during their daily activities. In addition to monitoring certain specific conditions, some have the added capability to trigger activities within the body. Prime examples are pacemakers, and pain management devices.
- Home automation: The list of smart home automation options includes: thermostats, electrical outlets and switches, doorbells, garage door openers, irrigation controllers, door locks, and security systems. Add to this a rather long list of entertainment and lifestyle management capabilities. Central to much

of these are home personal hubs from the likes of Google and Amazon. There are thousands of small apps that can be loaded on these devices to provide just about anything the user requests using their voice. And of course, just about all of these can be controlled from a smartphone.

- Smartphones: Packed with a bevy of embedded sensors, smartphones are seen as prime platforms for an ever-growing array of applications that use these sensors in various combinations to provide cool new capabilities and information for their users. Such applications can track where users have been, what they may want to know where they are, and where they need to go. Others monitor and report body vital measurements, like pulse rates, level of user activity, and the like. Smartphones also provide a personal remote control for smart home and entertainment devices.

4.2 AN EXAMPLE OF COST

At the heart of these solutions are sensors of all types, designs, and purposes, each with its own role to play in the search for more and better information that can be transformed and combined to yield new value. Sensors produce data—lots of it. And adding more sensors to a particular solution can add dramatically to its complexity. Each type of sensor is configured to produce a new observation instance of data many times a day. One sensor generating a 1Kb data record once each minute of each day will generate 1,440 records and 1,440 Kb of data. Doesn't sound like much. Many sensors generate records at least once per second. This becomes 86,400 records and 86,400 Kb each day. Depending on the environment and solution particulars, a significant portion of sensors may multiple observations per second. A hypothetical solution might have 100,000 sensors producing like data might yield 8.6 billion records and 8.6 GB of data per day. And if the hypothetical fully-rationalized solution ob-

servation processing cost is \$1 perMB, then the daily cost would be \$8,600 per day, \$258,000 per month, and \$3,096,000 annually.

Few solutions would be this simplistic. Typical solutions may include many different kinds of sensors, each producing observation data instances with specific content and size that is likely to differ from other solution sensor types. Each sensor will be configured to generate event observation data records appropriate to its particular need. In one solution, a specific sensor type might be configured to produce an event observation data record once per hour while another instance of the same sensor in a different application might be configured to produce one every second, yielding radically different record and data volumes each day. Some solutions may be deployed using a fixed local infrastructure while others may use wireless or satellite services. Choices like these will have much to do with transport and processing logistics, and ultimately, cost. Many solutions deploy 5-10+ times the number of sensors discussed in these examples.

sectionSensor-Based Solution Internal Controls While cost remains a consistent consideration in the overall IoT value proposition, data quality, processing latency, data volume, and overall solution performance are all critical factors that remain in a precarious state of balance. To ensure that IoT solutions deliver high data quality, they must include a strong system of internal controls aimed at preventing, detecting, and correcting defects or conditions that may compromise data quality.

4.2.1 PREVENTIVE CONTROLS

In the prior chapter, several sensor issues were identified and discussed. The discussion there established that the sensor world is young and still developing, and is yet to reach a point of maturity where data quality assurance can be reliably and consistently delivered at highly confident levels. Consequently, there are many exposures to the risk of poor data quality or loss. As a critical organizational asset, confidence

in data quality and integrity is essential for deriving the value the organization needs and expects. Preventive controls consist of various measures aimed at precluding the occurrence of undesirable outcomes and defects that may compromise the high quality of data. Such measures are usually more extensive and complicated than other types of controls, and invariably more expensive to implement and maintain.

Examples of preventive controls that could help reduce the impact of these might include:

- *Technical Limitations* - A preventive measure for technical limitations and some of the issues that arise as a result would be to make them more capable. This is entirely doable. But, sensors are designed and produced to be functional at the lowest cost—essential to justifying solutions financially. Increasing capacity, functionality, and capability all drive up the cost per unit. Multiplying that incremental cost by the number of units expected for a given solution can significantly change the financial equation. That said, manufacturers are finding new ways over time for sensors to do more at less cost. So, over time, sensor capabilities may become available that will help improve data quality.
- *Power* - Better power is a strong preventive measure for reducing data quality problems. Investing in cleaner, more stable building power, and good quality, long-life batteries would make an important difference when it comes to sensor function and performance. But this also can be expensive. Power systems in well-established facilities are what they are, and upgrading them can be cost-prohibitive, making it impractical to do. Batteries are getting better as the technology evolves, but upgrading 10,000 sensors to take newer battery models can be logistically difficult and will take time. And there is a cost to that as well.
- *Location* - Protecting sensors from environmental contamination might also be

an effective preventive measure for those devices where such action is possible. Alternatively, investing in components that are sealed or encapsulated in material that is impervious to the elements may be an effective preventative measure for prolonging the life of exposed sensor and sensor-related components.

- *Noise* - Noise is a complex and multi-faceted problem that can't be easily mitigated. However, a notable contributing factor is electromagnetic radiation. Some preventative measures can be taken to shield sensor and sensor components from such radiation. Machines can be shielded, network and electrical wiring and cabling can be relocated or shielded, and other sources can be traced to see if they can be mitigated. Such efforts can also be costly. Even so, stray signals and remaining interference can still cause problems.
- *Accuracy* - Drift is inevitable with most sensors and is rarely preventable. However, some data problems can be prevented by ensuring that the most appropriate sensor and sensor components are deployed and accurately documented so that they can be configured correctly to generate the intended data results and precision.
- *Duplication* - Options for preventing duplications are limited. Perhaps more capable onboard processing power may be able to at least minimize event observation record duplication, but probably not at this time.
- *Connectivity* - Investing in fault-resistant networks of sufficient capacity and speed will help prevent data streaming flow-control-related problems and their impact on data quality. However, upgrading existing networks can be quite expensive.
- *Fusion* - Fusion opportunities are specific to the targeted solutions and the role sensors or multi-sensor packages. Increasingly more of these are being built

with sufficient processing capacity to perform on-board fusion tasks. As these become more widely available may help prevent some of the fusion-related data quality problems.

From these examples, there is support for the idea that implementing certain preventive controls can mitigate some significant data quality exposures at significant cost. As in all internal control systems, such efforts should be considered only if there is sufficient cost-benefit, however that may be measured, for doing so. It is important to recognize that a solution may consist of thousands of locations and sensors. It is often impractical and too expensive to address all locations and devices. Areas where higher concentrations of data quality loss or impact can be identified, should be the better candidates for these kinds of remediations.

4.2.2 DETECTIVE CONTROLS

Central to all notions of data quality management is the core principal that data quality defects and issues should be able to be identified, measured, and assessed. Systems of internal detective controls are designed and implemented to ensure that this can be achieved as an intrinsic element of any ongoing IoT solution operation. This is particularly important in the absence of strong, robust, solution design and preventative controls. The generally accepted elements of internal detective controls include: environmental controls, risk assessment, monitoring operations activities, exchange of information, and assessment. For the purposes of this discussion, these elements will be adapted, for the purposes of this paper, to fit more appropriately the issues and characteristics of IoT sensor-based solutions.[24] [25]

ENVIRONMENTAL CONTROLS

Environmental controls focus on ensuring that the end-end environment and infrastructure used by any particular IoT solution is secure, functional, and properly con-

figured to create, process, move, and manage data flows that yield reliable, high quality data. Because sensor-based IoT solutions involve such large volumes of data, they need to operate effectively, efficiently, and reliably machine-to-machine with no human intervention needed to capture, clean, and prepare data for down-stream analytics and operational effectiveness measurement.

The foundational building blocks for solution environmental controls are assets: the assets deployed to support the solution, and the data assets the solution brings to life. To that end, such controls include the following:

- All devices are known and accounted for. Ensuring that when new sensors and related devices are brought online, they become known to all asset management, monitoring, and data management control processes.
- All changes to devices and the supporting infrastructure are fully coordinated. Formal change management practices are followed when changing anything about a solution element that could impact data quality.
 - *Sensor-affiliated devices:* Sensors, actuators, controllers, processors, and data processing computers are all considered assets and have specific roles to play within the IoT solution they support. Each needs to be identified and known to the network and to solution management and control functions. Once online, their status, health, and performance needs to be monitored and measured. As solutions evolve, devices break or begin to malfunction, devices need to be expanded or upgraded, or even retired. In each of these cases, needed changes or updates need to be coordinated and tracked, and confirmed back online and functioning properly when the actions are completed. A fully functioning infrastructure supporting an IoT solution is an essential foundation for dependable data quality.

- *Data formatting rules:* Every solution device that creates or further processes data does so according to a specified set of data format rules. All devices of a specific type are expected to adhere to the set of data formatting rules intended for them. As solutions evolve and mature, data format rules expand or change and are assigned to pending or future updated rule sets. In a coordinated fashion, the current set is retired and replaced by the next designated set. At that point, all impacted devices should begin complying with the new set. This should help ensure that any user or process using that data does so according to the correct set of rules. This is another essential part of the foundation for dependable data quality.
- *Validation rule set versions:* Like data formatting, most created data records have a set of rules for validating that the data within them are correct, complete, and have values within specified ranges, or that are from lists of acceptable values. Like formatting rules, these are updated from time to time and should also be managed through version control and change management processes.
- *Machine learning algorithms:* Machine learning models and algorithms are being used increasingly for data cleaning, enhancement, and transformation in IoT solutions. Many sensor-originated data streams may employ ML processes for resolving data issues. Where they are used, all data streams from the same device type should use the same version of models and/or algorithms. As they are refined and updated, they should also be managed through version control and change management processes.

RISK ASSESSMENT

In any IT solution, security and privacy protection are critical and essential elements of application solutions. IoT solutions by nature have many parts, each of which

should be expected to provide protections consistent with those provided by all other solution components, and what the overall solution should be expected to provide overall. As in any chain, the solution is only as strong as its weakest component. Due to the general immaturity sensor-based IoT solutions, there remain many exposures and risks that can negatively impact data quality, security, and privacy.

In 2018, the Open Web Application Security Project (OWASP) Internet of Things working group published its list of Top 10 IoT security vulnerabilities **owasp**. They describe them this way:

1. *Weak, guessable, or hardcoded passwords:* - “Use of easily brute-forced, publicly available, or unchangeable credentials, including back-doors in firmware or client software that grants unauthorized access to deployed systems.”
2. *Insecure network services:* - “Unneeded or insecure network services running on the device itself, especially those exposed to the Internet, that compromise the confidentiality, integrity/authenticity, or availability of information or allow unauthorized remote control.”
3. *Insecure ecosystem interfaces:* - “Insecure web, back end API, cloud, or mobile interfaces in the ecosystem outside of the device that allows compromise of the device or its related components. Common issues include a lack of authentication/authorization, lacking or weak encryption, and a lack of input and output filtering.”
4. *Lack of ability to securely update the device.* - “ This includes lack of firmware validation on device [sic], lack of secure delivery (non-encrypted in transit), lack of anti-rollback mechanisms, and lack of notifications of security changes due to updates.”

5. *Use of insecure or outdated components:* - “Use of deprecated or insecure software components/libraries that could allow the device to be compromised. This includes insecure customization of operating system platforms, and the use of third-party software or hardware components from a compromised supply chain.”
6. *User’s personal information misused:* - “... stored on the device or in the ecosystem that is used insecurely, improperly, or without permission.”
7. *Lack of encryption or access control:* - “... of sensitive data anywhere within the ecosystem, including at rest, in transit, or during processing.”
8. *Lack of device management:* - “Lack of security support on devices deployed in production, including asset management, update management, secure decommissioning, systems monitoring, and response capabilities.”
9. *Insecure default settings:* - “Devices or systems shipped with insecure default settings or lack the ability to make the system more secure by restricting operators from modifying configurations.”
10. *Lack of physical hardening:* - “Lack of physical hardening measures, allowing potential attackers to gain sensitive information that can help in a future remote attack or take local control of the device.”

Vulnerabilities like these expose IoT solution elements and data to risks of hacking, intrusion, and compromise that in the end, could result in losses significantly greater than the solution cost; they need to be addressed. Every location and every device in each location needs to be assessed and hardened. Nearly every sensor creates a data channel that needs to be reviewed as well to find potential weaknesses. This is a big topic with many nuances that underlie nearly every other facet of the IoT issues and solutions discussed in this paper. The topic needs to be discussed in much greater

detail in other papers and won't be discussed further here. This does not lessen the concerns and exposures discussed here.

MONITORING OPERATIONS ACTIVITIES

The goal of all monitoring activities is to ensure that all data is flowing as expected at all times throughout the universe of IoT solution elements. Monitoring capabilities should be incorporated into solutions to inquire, test, confirm, and record in a log the status of all monitored elements. Every element should be observed on a regular schedule appropriate for the role that element plays in the overall solution. All elements in the same role are likely to follow the same schedule. The frequency of observation is likely to be a function of several factors like volume, role critically, time sensitivity, degree of volatility, potential exposure to loss or error, and the probability of risk that such a loss might occur. High volumes of critical time-sensitive data may need to be observed every minute, while at the other extreme, low volumes of non-critical, low volatility, low risk data might be observed hourly.[11] [14]

It is easy to imagine that if each of the monitoring observations generated data records of their own that such volumes could be very large and could dramatically compound the total volume of data produced by a particular IoT solution. For this reason, monitoring would better take a "dashboard update" approach to logging. With this approach, it is assumed that all is well throughout and it is sufficient to simply track the most recent time of observation. So, other than the dashboard attributes representing a particular solution element, no new data is generated for positive observations.

However, it is rare that everything runs smoothly all the time; there are usually some number of conditions or defects that need further attention and need to be documented and tracked until they are resolved. Once discovered, these need to be appropriately acted upon per processes prescribed in the category of corrective

controls which are described in a later section in this chapter.

As stated, operations monitoring activities focus on discovering anything within the IoT solution that may be considered anomalous or outside stipulated rules and specifications. The methods for making such discoveries can sometimes be complicated and may be approached in a variety of ways. This is purposely left to the solution designer to determine the best means of making these discoveries.[27] The intention here is to suggest the general principles and objectives that might be used to pursue such efforts. Examples of activity monitoring actions might include these.

- All expected data from all expected sources is present for all expected in-service times. If there is a break or a gap in a particular data stream, which is it?
- All data is formatted as expected and complies with the current version of validation rules and algorithms. Have current rules and templates been applied to correct most data errors and defects coming from sensors? What else may need to be looked at for further correction? Such might include:
 - noisy data
 - duplication
 - missing or inaccurate attribute values
 - outliers
 - missing or broken records; records at random; in blocks; observed vs. unobserved
 - evidence of sensor drift
 - sensor & sensor data fusion errors
- Reliable file routing. Every solution sensor is the head-water of its own datastream that has a defined destination. At one or more process stops along the

way, data records need to be validated, formats need to be checked, and identified defects fixed. Certain record attributes may be cross-validated with other record attributes from other streams passing through the same process stop. Many data streams are ultimately merged with others, some early on and others further down stream. Each stop is a designated computing venue in the IoT datasphere described in an earlier chapter. Low latency, low-moderate volume, and low processing demand data streams will most likely be processed in the closest computing venue to meet solution requirements. All other data streams will be mapped appropriately to computing venues further down stream.[10] [12]

There are three primary control functions here that need to be incorporated.

1. *Planned routing verification.* Confirm that each data stream is flowing from creation to destination as expected and on time. It is essential to know that all is running smoothly. Breaks or gaps in data flows need to be identified and investigated. Potential issues or losses may occur if data isn't arriving where it is expected to be when it is expected. For example, a data stream from a sensor driving a process control function has a very narrow window in which to fuse with data from other nearby sensors, clean up the data, derive a collaborative reading, and send those results to an actuator that may need to make an adjustment accordingly to the process flow. If those results are late, or don't arrive at all, adjustments may be made at the wrong time or not at all. The process batch might be ruined or become defective as a result. Events like these need to be localized quickly and investigated for resolution, and tracked until normality is restored.
2. *Routing exceptions.* Most comprehensive solutions are designed to anticipate the possibility that planned primary data stream routes may fail.

Edge or Fog hubs or gateways are inserted at key transfer points to manage traffic paths that host the various data streams. When the planned routes become inactive, those hubs or gateways select alternate routes for the data streams that are affected. These events create routing exceptions that need to be tracked, monitored, investigated, and remediated to restore the planned routes and move data streams back to them. For example, a local wide area network route may fail and be rerouted to a 5G or satellite route to bypass the downed primary route.

The fundamental issue this creates is the abrupt disruption in the data streams impacted. They will no longer be present on the downstream legs of the primary route, so they will need to be tracked as issues until resolved. Notifications are needed to ensure that everyone that needs to, knows that these routing exceptions are in effect. New temporary routes need to be built and managed dynamically to look for those data streams at the hub or gateway at the edge venue of the new route provider which could be considerably greater distance away from the next normal computing venue that will be looking for those data streams.[22] Once everything is reconnected, a synchronization process is needed to reconcile continuity issues such as gaps, overlaps, duplications, and corrupted records or blocks of records. Events like these can be complicated and require full immediate attention until all is back to normal using planned routes. Exposures and risks are high in these circumstances that data quality and integrity can become corrupted and compromised.

3. *Routing performance metrics.* Route planning is a critical operational management function for an IoT solution of scale involving multiple sites, multiple service providers, and multiple technologies. Most solutions do. A solution with 10,000 sensors across 100 US locations will start with about

10,000 data streams. Some of those data streams will be processed locally, fulfill their purpose and be retired. Others may do the same but also forward data to downstream computing venues in the fog, or the edge, or on to one or more cloud service providers. Many data streams will be fused or merged with others along the way, reducing considerably the number of streams that will ultimately be inserted into the solution data stores. Data stream routes could interconnect with a variety of technologies and services for which the solution will need to keep track.[15]

How this is accomplished is very much a function of the solution itself and the requirements it is designed to meet. What is essential to this function is to be able to identify and track data streams that are losing data, or processing slower than planned, may be making observations too frequently or not often enough, and other conditions that may negatively impacting data stream flows, processing performance, route throughput, and ultimately data quality and integrity.

4.2.3 CORRECTIVE CONTROLS

In this discussion of internal controls for IoT solutions, a variety of possible defects, errors, problems, and exceptions [collectively referred to as issues] have been identified. Each has potential negative impacts on solution integrity, operations, performance, and data quality. The system of detective controls is designed to identify any such issues, and where within the solution the issue appears to be occurring. The purpose of the system of corrective controls is to establish and follow processes to fix those issues. Simply put, corrective controls take in identified issues, characterize and understand them, determine what is causing them and why (root cause), ascertain where they are occurring, develop a plan for fixing them, execute the plan to fix them, and confirm that those actions have corrected what was identified.

What exactly is the issue? The answer to this question together with the information gathered to help support that answer needs to be clearly understood before doing anything else. Questions that may need to be considered include: How can it best be described? Is this a common issue? How frequently does it occur? When, where, and under what conditions did it last occur? Under what conditions has it occurred most often? Is there enough information to allow issues with similar characteristics to be excluded from further investigation?

What is the root cause? Finding the root cause(s) behind an issue is fundamental to developing a clear understanding of the factors that may be causing it. Particular care is needed to distinguish symptoms from causal factors. Discovering the true root cause(s) of an issue is a disciplined process seeking answers to questions that are driven usually by the known or suspected issue circumstances.

Where is it occurring? Once the root cause(s) of an issue are understood, other locations and/or impact points are investigated to determine where else the issue is appearing so that they can also be fixed. Patterns often emerge during earlier analysis that may point to like issues, even if they are not showing issues at the time. For example, one particular sensor make and model in multiple locations might be showing similar issues intermittently over time and the root cause findings suggest that each is experiencing the same issue with a threshold limit parameter value in its configuration table and needs to be remediated. In another example, the root cause findings uncovered multiple related issues from several devices that happen to be connected to the same hub. The hub was investigated and found to have an unstable network interface card.

What is the plan for fixing it? Once it is clear what needs to be done to what to fix the issue, a change management plan outlining the specific change steps and schedule for each device and location. The plan should then be reviewed and approved by all impacted stakeholders.

Follow the plan to fix it. At the appointed times agreed between operations, change managers, other interested stakeholders, the changes are made exactly as outlined in the plan in full coordination to minimize operating impacts. Plan steps should be included to ensure that all related documentation and metadata are updated to reflect the changes made.

Is the issue fixed? Part of that plan should include for each device and location the steps to be taken to confirm that the changes were made correctly and they have effectively remediated the identified issues and demonstrate the intended outcomes have been achieved.

CHAPTER 5

CONCLUSION AND FUTURE WORK

The entire concept of the IoT is exciting and will provide fertile ground for innovative new development for at least the decade ahead. Time and creativity over time will help spawn evolving ideas about using IoT-supporting sensors to solve important problems, find better versions of today's solutions, and do today's things in better ways, will all drive the creation of a host of new applications and their variant solutions and more data about virtually anything. However, the drive for these new benefits cannot come at the risk of compromises and weaknesses in data security, personal privacy, organizational intellectual property, and customer information.

As mentioned several times herein, all this is about data, lots of it. Even now, there is so much of it that a substantial portion of what Gartner calls "dark data" never goes anywhere or is ever used. Further, data needed for analysis must be discovered and extracted from these mountains of potentially redundant and nonessential data. A lot of data has a brief, useful life and may need to be extracted and destroyed rather than kept. More attention needs to be directed toward where data is created and where it needs to be processed or analyzed. Moving large amounts of data long distances across the network is expensive, time-consuming, and may block others from using the network when needed, particularly when low-latency response and or immediate processing is required.

Navigating the layers of infrastructure supporting the IoT is a big topic with many facets that this paper could not adequately discuss. However, it needs to be evaluated closely regarding the data crisis that, in many ways, is already upon us.

5.1 THESIS QUESTIONS AND CONCLUSIONS

This thesis aims to discuss the issues surrounding sensor-generated data volumes and data quality and find answers to the questions posed in the Introduction. Here then are the findings and conclusions to those questions.

1. *Given the increasing magnitude and velocity of sensor-generated data being produced, transported, and stored, the question must be asked: "is all this data necessary?"* No! Not all collected data is necessary, nor does all data need to be collected. And not all collected data needs to be kept. There should be a direct correlation between the data that is collected, processed, and stored; and the specific requirements that require it to enable the creation and ongoing production of new value targeted by senior organizational leadership. The literature generally agrees that too few organizations have clearly defined what added value they seek and the specific data needed to achieve it. Without such clarity and focus, the default action then is to, generally, capture all data and try to work out the value opportunities later. While this may be a practical approach for certain kinds of value exploration, it is categorically not appropriate for all situations.

The academic and trade literature identify this focus of action-to-value concept as one of the central facets of sound data governance. But, unfortunately, the typical organizational adoption rate is slow and modestly penetrated. Moreover, implementation maturity and degree of compliance across most organizations are considered low. Consequently, such organizations have not yet reached a point where they clearly develop their value propositions and exercise the discipline to identify the specific data needed to achieve it.

So, infrastructure providers continue to expand their capabilities to improve their cost-efficiencies in the ways they transport, process, and store all the data

their clients choose to generate and retain. And at current and projected rates, there are mountains of data to come for the foreseeable future.

Meanwhile, middleware and application solutions providers are evolving their wares to improve their clients' ability to manage better the data they have and the new data they create every day.

2. *How much of this data is actually being used?* No one seems to know for sure. Estimates in the literature and industry sources vary widely, offering little consensus on precise numbers—the data just isn't tracked or recorded consistently to scale. In 2018, McKinsey analysts estimated that less than 4% of sensor-generated data is used, primarily for anomaly detection and control. The higher value uses of this data for optimization and prediction is limited and just beginning to be explored. Industrial and business-to-business (B2B) applications appear to offer the greatest opportunities for value creation and growth, followed by consumer-focused applications. The scientific data community is increasingly adopting the IoT model as a part of its overall data management transformation.[23] In so doing, it is expected to use a significantly larger portion of data collected, although small by comparison. Another higher use area is the green-field deployment of IoT application solutions in developing economies where value-connected data collection and retention is expected to be more focused.
3. *What strategies can be applied to materially reduce data volumes and their attendant resource demands and costs?* In retrospect, the issue is not simply reducing data volumes as much as it is collecting, processing, and retaining only data that is of sufficiently high quality that it can be used readily to produce the value for which it is intended. As discussed previously, the following support this premise:

- Pragmatically, only data that is needed and is likely to be used to achieve a specific purpose should be captured. Incurring all related costs for data that isn't likely to be used wastes time, money, and resources.
- Data may be deleted or archived appropriately when its purpose has passed. However, if it needs to be retained, perhaps it can be done at a lower cost.
- Data may be collected more frequently than needs require. Lowering frequencies produce less data overall and reduce associated costs and resource demands accordingly. Of course, the opposite may need to be applied as circumstances dictate. If the frequency of observations is insufficient to satisfy requirements, they may need to be increased.
- Data may be useful at multiple levels and needs to be managed accordingly. For example, data may be captured and used immediately to represent the precise reading of a real-world circumstance and then evaluated by a proceeding process as to whether or not a related actuator needs to make an indicated adjustment as a result. If this is the only purpose for that data, it may not need to be processed further downstream and deleted. However, if that data instance also has analytical or predictive value, it should be forwarded downstream and retained.
- In many cases, consecutive data instances may show the same values with no appreciable or material differences. There may or may not be merit in processing and retaining these instances. In these situations, it might be appropriate to drop them or design solutions to summarize these consecutive series of inconsequential observation data instances and devise methods for retaining and analyzing them.

4. *What data quality challenges are a result of environmental and operational in-*

fluences on sensors supporting IoT? The past two chapters describe the nature of sensor issues that can impact data quality in IoT application solutions. In their current form of production and availability in the low-end cost range, they have limited abilities to operate reliably in harsher environments and cope with a wide variety of potential operational influences. However, the author's research found no recent broad empirical numbers indicating the magnitude of such errors on a regular basis. That said, the incidence of these errors tends to be circumstantial. The most common sensor-related data issues are outliers, followed by missing data, bias, drift, and noise. Less frequent issues include constant values, data uncertainties, and stuck-at-zero.

In addition to their individual functional requirements, application solution developers must also anticipate and resolve these errors. Further, the IoT environment has a host of security, privacy, and access control challenges that may impact data quality and reliability; have not been researched as part of this paper. Solution developers must incorporate needed controls and functionality to mitigate as many of those challenges as circumstances allow.

5. *What methods can be introduced or expanded upon to improve IoT data quality?*

Data is most useful when it is of sufficiently high quality that it can be trusted and is believed to reliably represent the real-world conditions accurately enough to satisfy its intended purpose. When the level of data quality falls, so does the trust and confidence in the data itself. At some point, it is considered no longer useful and is either deleted or ignored. The issue here is that this is (hopefully) data with a purpose. When it is deemed unusable, it creates a gap, be it a single instance or block(s) of many data instances, and removes observations from the data stream that are potentially consequential. While the literature identifies various methods that may be applied to compensate for what has been lost, it still may not be what was originally observed.

As outlined in the previous chapter, the ability to detect potential errors is the essential starting point for evaluating and performing appropriate corrective actions needed to mitigate, as much as possible, the impacts such errors may have on the integrity and reliability of individual sensor data streams. Each type of error may lead to a specific course of correction and resolution and may differ further depending on the sensor make and model creating the error. Real-time error detection and correction can be complicated because of the limited time window within which such actions can occur. These typically leave little time for time-consuming statistical or algorithmic solutions, while data streams with no immediate processing demands can benefit significantly by applying them.

In real-time conditions, few options may be available. In these conditions, time may only permit the application of basic attribute pattern matches of various types to determine if an error may exist. Options may include simply ignoring the erroneous instance(s) or substituting a statistically-derived Proforma replacement to satisfy the real-time need. Faulty data instances are usually retained in the data stream so that they may be included in downstream detection and correction processes.

A wide variety of approaches are suggested in the literature for detecting and correcting defective/erroneous/anomalous data. Common methods include principal component analysis, artificial neural networks, Bayesian networks, Grey prediction modeling, benchmark-based calibrations, Kalman filters, signal processing and filtering, association rule mining, clustering, k-nearest neighbor, and value/mode decomposition. Depending on how they are used, some are more suitable for detection, while others may be better for correcting or replacing missing data. Even further, some are more suitable for certain types of errors or defects, or even specific types or categories of sensors.

As described previously, sensors have specific characteristics and purposes. The data produced by each sensor make and model are usually distinct in the attributes captured and what values they should contain. Precise detection and processing methods can be established and routinely applied to all data streams originated by them. Benchmarks and profiles can be derived over time from the community of similar sensors helping to improve detection, correction, and data quality.

5.2 FUTURE WORK

Opinions vary broadly about how far things may be in developing and deploying sensor-based application solutions. Many would probably agree that applications, their assorted solutions, and underlying technologies are emerging from the "Early Adoption" stage into the (very) "Early Majority" stage of the Technology Innovation Life Cycle, and the low end of the steep "Improving" slope of the Technical Maturity Model with "Maturity" still a long way off. In many ways, IoT technology and applications still fall far short of the inflated expectations and hype of the marketplace. The IoT is expanding on an extraordinary and accelerated scale regardless of the current state of everything that must be there to achieve its ultimate potential. Through this research effort, several key critical success factors were identified that will need concerted industry focus to be achieved. These include the following:

1. Security, access, and privacy protections must be incorporated at all levels of every solution and infrastructure element it uses to reach the levels of reliability, confidence, and trust that are essential to realizing its full intended value and innovative potential. There are far too many serious weaknesses and vulnerabilities today across all of the actors engaged in the development, deployment, and operation of any given solution and are collectively negligibly "seamless." The entire burden currently lies on the shoulders of solution providers and develop-

ers to mitigate whatever they can or so choose. They will carry this burden, expense, and liabilities until something much more broadly functional is available that they can use. Industry remedies need to be implemented as widely as possible using universal standards that all can adopt and support. IoT infrastructure and solution security and privacy weaknesses are critical problems and must be fixed. And it will take an enormous amount of effort over time to get there. All other observations that follow are predicated on the presumption that this must and will happen and will not mention it further.

2. One of the underlying visions of the IoT is dynamic machine-to-machine(M2M) interoperability, the notion that each device is independent, self-managing, and self-determining.[20] Each would seek out other devices and determine which it should establish relationships with and which routes it should use to get its data to where it needs to be. Such interoperability has been on the horizon for many years, and many now see the IoT as a primary vehicle of innovation for making it a reality. Others believe, however, that too much development and deployment has been done "today's way" to make this even marginally possible over the next several years. Moreover, there have been virtually no industry-wide efforts to standardize the essential transport, processing, and data hygiene functionality and protocols that will be needed to enable even "baby steps" in the direction of this lofty objective.
3. IoT connectivity is problematic today and represents solution-impacting limitations and challenges. Inconsistencies in coverage, bandwidth, and quality of service (measuring throughput and error rates) across a given solution service area make it difficult to ensure that data reliably gets to its designated destination. For example, selected solution elements targeted for certain wireless services can't be supported in some areas because the needed wireless service

is not offered. As a result, some other form of connectivity may be required. In some cases, it may be through an alternate provider. If data streams are intended for the primary provider's Edge data centers, alterations may be needed to accommodate differences between the two provider Edge service offerings. A bridge between the participating Edge centers will need to be built to facilitate data stream controls and flows.[10] If the alternate provider can only provide lesser service levels, then error rates may be higher, latency may be higher, etc., adversely impacting data quality and solution performance. Several examples of related problems will need further research and resolution if the IoT is to become what it needs to and can be. Each variant adds complexity and added effort to make it all work.[15]

4. Every device that has a role in the ongoing operation of an IoT solution needs to be known and managed under a system of solution controls for various important reasons. Each device is an asset that must be tracked and managed according to rules and processes applied to all other assets. Management features include applicable periodic maintenance and calibration, software version control, physical location tracking, role assignment metadata updates, ongoing performance metrics and service issues, support organization contact details, other devices that it talks to, and the like. If it exists at all, the extent of this functionality is a function of how the solution using it was designed. Establishing and maintaining this level of control is essential for operational reliability and service management that begins to get very complicated when more of these devices are shared by multiple solutions touched by multiple solution service providers. The maturity and functionality of service management operations capabilities will become a much bigger issue as industry providers, and solution support organizations begin to drive towards ubiquitous M2M interoperability. A universal service-layer offering will be the essential foundation through

which all solutions can manage what will become a very dynamic infrastructure landscape.[19]

For example, a local hub supporting multiple wired sensors loses its connection to a downstream Fog device. The hub searches for possible routes that will restore its path to nodes further downstream. An alternate path is found through a 5G wireless connection through the provider's Edge data center to the closest node in the original data path. All this will need to occur dynamically and automatically and will need to update all impacted device metadata and control processes. All stakeholders of all devices and impacted solution elements will need to see what they are authorized to see and continue monitoring and managing activities with the dynamic change forward.

5. As described in the previous chapter, each data stream needs to be managed as it flows from its point of origination to its intended destination. A system of controls needs to be incorporated into solutions to monitor and track each data stream along its current path. Such controls ensure that all data from all sources are present and accounted for and that variances that occur along the way are adequately documented.[11] In most of today's solutions, material changes in devices, their status, and data paths are managed manually, some better than others. Every break or change in a datastream's path or flow must be handled as smoothly as possible to minimize potential data loss or error. Changes impacting low-latency data flows can break operational integrity if the latency in the new path rises above critical latency thresholds. These processes will also become much more complicated as device management becomes more dynamic, as previously just described.[12]

As in the previous example, changes in device status and connectivity also impact datastream monitoring and control processes. Changes in datastream

paths will need to be instantly and dynamically logged to minimize the disruption of visibility to the flow and processing of the impacted data along its new course. Similarly, there will be a need for a universal datastream flow control layer that can provide all authorized stakeholders the access and control capabilities they will need to manage what they are allowed to do. Building such a capability will take a major commitment of providers, developers, and other stakeholders to develop the standards, protocols, and tools needed to make all this possible.[13]

6. And last but not least, cost-sensitive sensor evolution and innovation need to proceed forward as rapidly as possible. Critical to this effort is developing designs that bring increasing memory, computing, storage, and data flow control on-board the sensor package or be readily adjacent. Incorporated into resident capabilities should be self-diagnostic and error-detection functionality to preclude the need for downstream evaluation, particularly in real-time, low-latency situations. In addition, self-healing and recalibration, and the reporting of suspected issues that be picked up downstream by monitoring processes. Finally, device access security provisions need to be added and made manageable.

Sensor fusion management capabilities need to be added and expanded to facilitate the interconnection and fusion of data originating from adjacent content- and context-related sensors. As sensors evolve to manage more of these functions at internal low-latency speeds, responsibility for similar solution functionality can be taken off the shoulders of solution developers. Advancing functionality and capabilities aimed at improving data quality will challenge sensor design engineers to innovate in ways that keep sensor costs as low as possible.

5.3 EPILOGUE

IoT solutions are complicated with many moving parts. Some solutions span multiple geographic areas with large numbers of many different kinds of sensors tasked to function in widely-varied operating environments using locally-available infrastructure service options and providers. As long as developers must cobble together solutions consisting of so many unique moving parts from multiple vendors and providers, the task of developing world-class, high-reliability solutions yielding consistently high data quality will demand an increasingly broad palette of skills, patience, money, time, and most importantly, senior leadership support. Until the industry can successfully develop and deploy standards, protocols, and enabling tools for common cross-platform services, today's problems will continue to expand and become more challenging to manage. The volume and velocity of device and data growth into the foreseeable future may ultimately cripple or collapse, sooner than anticipated, solutions being deployed today. New approaches will be needed that is entirely woven together with industrial-strength, bullet-proof, security, and privacy protections. All manner of participating stakeholders will need to collaborate heavily to define and implement the required standards and capabilities— *much sooner than later!*

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